

Russell Dudek

Industrial Systems Modernization | Manufacturing Data & Automation

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<https://russelldudek.github.io/ByteCubit/>

Industrial Automation Engineer | Manufacturing Data & MES Modernization

Operator-technologist connecting production behavior, applications, databases, telemetry, and governed change.

EXECUTIVE FIT

Manufacturing systems leader with direct experience across high-reliability electronics, software-enabled robotics, quality systems, ERP workflows, telemetry, SQL, Python, standard work, and live operating environments.

Builds visible mechanisms for dependency mapping, root cause, release discipline, human authority, and measurable operating outcomes. Does not claim unverified production Ignition tenure or AVEVA PI administration.

ROLE-ALIGNED CAPABILITIES

Manufacturing systems and MES modernization | SQL and data workflows | Telemetry and machine-fleet signals

High-reliability operations | Quality systems and root cause | Release, rollback, and support mechanisms

Cross-functional work with production, engineering, quality, support, R&D, vendors, and customers

Python, JavaScript, Tableau, ERP/Odoo, Jira, Confluence, HubSpot, Slack, agentic AI and human review

VAPE-JET | DIRECTOR OF OPERATIONS | 2025-PRESENT

- Leads AI-first operating transformation across production, purchasing, quality, customer success, support, engineering issue flow, ERP/Odoo, Jira, HubSpot, Slack, telemetry, and AI-assisted workflows.
- Connects factory, field, customer, and machine-fleet signals into prioritized work, earlier risk detection, cleaner handoffs, dashboards, standard work, training, and accountability.
- Works directly in a software-enabled robotics and machine-vision environment with approximately 1,000 fielded robots informing support and operating feedback loops.

COMPUNETICS | DIRECTOR OF OPERATIONS AND ENGINEERING MANAGEMENT | 2000-2013

- Progressed through R&D program management, production management, strategic acquisition, operations management, and director-level leadership.
- Led engineering, manufacturing, quality, customer delivery, and complex technical programs in advanced electronics and high-reliability environments.
- Built quality systems, standard work, root-cause practices, performance measures, and process controls; collaborated with DoD, CERN, Intel, and AMD.

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AMAZON | OPERATIONS MANAGEMENT / SITE LEADER | 2014-2018

- Led launch and execution for Amazon Prime Pittsburgh in a 24/7 customer-backward operating environment supporting approximately five million second-day orders annually.
- Built leadership routines, labor planning, throughput visibility, escalation paths, standard work, and operating mechanisms; applied Lean and Gemba across safety, quality, throughput, customer experience, and engagement.
- Served on the Amazon Prime Now launch team with Whole Foods and contributed to Prime Air research and Amazon Flex geolocation concepts.

DUDEWORTH | FOUNDER / AI AND AUTOMATION ADVISOR | 2017-PRESENT

- Develops AI-readiness and opportunity frameworks covering value, feasibility, data readiness, governance, adoption burden, reuse, and time-to-value.
- Designs agentic workflow patterns using structured context, retrieval, human judgment, escalation, decision records, evaluation, and governance rails.
- Advises on AI-enabled operations, warehouse automation, AMRs, ASRS, cobots, transportation systems, network analysis, and workflow design.

ADDITIONAL EVIDENCE

Cardinal Building Products | Regional Director | 2021-2023

Built the company's first remote regional operation and modernized CRM, fulfillment, logistics, digital marketing, and online-buying workflows.

ZeusVu | Chief Pilot, Autonomous Systems and Aerial Data Operations | 2018-2022

Led regulated drone operations with a 100% safety record and no FAA incidents; developed TensorFlow-based concepts for automated aerial inventory and governed delivery.

IEEE | CMPT and EDS Pittsburgh Chapter Chair | 2012-2016

Led regional technology programming connecting industry, academia, electronics, manufacturing, and emerging technology.

EDUCATION AND CREDENTIALS

Bachelor of Science, University of Pittsburgh - academic grounding includes materials science and physical chemistry.

Google AI Essentials | Food Safety Management Certification | IBM Enterprise Design Thinking Practitioner | Six Sigma Certification | OSHA 10

IMMEDIATE CONTRIBUTION

Read the production system, map legacy MES/SQL/historian dependencies, establish a bounded Signal Twin acceptance record, and prove one reusable modernization path without overstating platform tenure.